YIHAN WANG

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EDUCATION

Ph.D. in Physics — 2017 - 2022
Stony Brook University, NY
Advisor: Rosalba Perna

M.A. in Physics — 2016 - 2017
Stony Brook University, NY

B.S. in Physics (Theoretical Physics) — 2011 - 2015
University of Science and Technology of China (USTC)

PROFESSIONAL APPOINTMENTS

Postdoctoral Research Associate — 2025 - Present
University of Wisconsin, Madison

Fellow, Nevada Center for Astrophysics — 2022 - 2025
University of Nevada, Las Vegas

RESEARCH INTERESTS

Dynamics of Extreme Environments & Multimessenger Astrophysics

My work combines theoretical dynamics with high-performance simulations to study the universe's most violent events. I focus on bridging the gap between small-scale interactions and large-scale observable transients.

- **Gravitational Wave Sources:** Formation channels of binary black holes in Active Galactic Nuclei (AGN) and dense star clusters.
- **Tidal Disruption Events (TDEs):** Hydrodynamical simulations of stellar disruptions and modeling their electromagnetic signatures.
- **Exotic Planetary Systems:** Origin of free-floating "rogue" binary planets and the stability of planetary architectures in chaotic environments.
- **Computational Astrophysics:** Development of high-precision N-body algorithms (e.g., `SpaceHub`) and rapid modeling frameworks for GRB afterglows (e.g., `VegasAfterglow`).

GRANTS & COMPUTING ALLOCATIONS

ACCESS-SDSC | *1.25M CPU hours* — 2023 - 2025
Project: *Propagation of Fast Radio Bursts* (PI)

ACCESS-SDSC | *0.5M CPU hours* — 2022 - 2024
Project: *Hot Jupiter Formation in Dense Star Clusters* (PI)

HONORS & AWARDS

Gerald Brown Prize	2022
Di Tian Prize for Excellent Research	2020
National Endeavor Scholarship	2014
Outstanding Student Scholarship (USTC)	2012, 2013
Exceptional Prize, Physics Experiment Contest (USTC)	2013

PUBLICATIONS

A complete and updated publication list is available on NASA ADS and Google Scholar.

[30] **Wang, Y.**, Chen C., Zhang B., *VegasAfterglow: A High-Performance Framework for Gamma-Ray Burst Afterglows*, 2025, submitted to JHEAP, arXiv.2507.10829

[29] Chen C., **Wang Y.**, Zhang B., *X-ray Emission Signatures of Neutron Star Mergers*, 2025, submitted to ApJ, arXiv.2505.01606

[28] Li W.-X., ..., **Wang Y.-H.**, ..., *An extremely soft and weak fast X-ray transient associated with a luminous supernova*, 2025, submitted, arXiv.2504.17034

[27] **Wang Y.**, Perna R., Zhu Z., Lin D. N. C., *Evolution of Extremely Soft Binaries in Dense Star Clusters: On the Jupiter Mass Binary Objects*, 2024, submitted to ApJ, arXiv.2310.00020

[26] **Wang Y.**, Graham M., Ford S., McKernan B., Ryu T., Stern D., *Conditions for Changing-Look AGNs from Accretion Disk-Induced Tidal Disruption Events*, 2024, submitted to ApJL, arXiv:2406.12096

[25] **Wang Y.**, Zhang B., *Evidence of a Past Merger of the Galactic Center Black Hole*, 2024, Nature Astronomy

[24] **Wang Y.**, Perna R., Zhu Z., *Floating binary planets from ejections during close stellar encounters*, 2024, Nature Astronomy

[23] **Wang Y.**, Lin D. N. C., Zhang B., Zhu Z., *Changing-Look AGN Behaviour Induced by Disk-Captured Tidal Disruption Events*, 2023, ApJL, 962, L7

[22] Prasad C., **Wang Y.**, Perna R., Ford S., McKernan B., *Tidal Disruption Events from three-body scatterings in the disks of Active Galactic Nuclei*, 2023, submitted to MNRAS, arXiv.2310.00020

[21] **Wang Y.**, Zhang B., Zhu Z., *Anisotropic Energy Injection from Magnetar Central Engines in Short GRBs*, 2023, MNRAS, 528, 3705

[20] **Wang Y.**, Zhu Z., Lin D. N. C., *Stellar/BH Population in AGN Disks: Direct Binary Formation from Capture Objects in Nuclei Clusters*, 2023, MNRAS, 528, 4958

[19] **Wang Y.**, Ford S., Perna R., McKernan B., Zhu Z., Zhang B., *Effective two-body scatterings around a massive object*, 2023, MNRAS, 523, 2014

[18] Xin C., Haiman Z., Perna R., **Wang Y.**, Ryu T., *Tidal Peeling Events: low-eccentricity tidal disruption of a star by a stellar-mass black hole*, 2023, ApJ, 961, 149

[17] **Wang Y.-H.**, Lazzati D., Perna R., *The emergence of diffused Gamma-Ray Burst afterglows from the disks of Active Galactic Nuclei*, 2022, MNRAS, 516, 5935

[16] Ryu, T., Perna R., **Wang Y.-H.**, *Close Encounters of Stars with Stellar-mass Black Hole Binaries*, 2022, MNRAS, 516, 2204

[15] Perna R., Artale M. C., **Wang Y.-H.**, et al., *Host galaxies and electromagnetic counterparts to binary neutron star mergers across the cosmic time*, 2021, MNRAS, 512, 2654

[14] **Wang Y.-H.**, Perna R., Leigh N. W. C., Shara M. M., *Hot Jupiter formation in star cluster: Secular chaos*, 2021, MNRAS, 509, 5053

[13] **Wang Y.-H.**, McKernan B., Ford S., Perna R., Leigh N. W. C., Mac Low M.-M., *Symmetry Breaking in Dynamical Encounters in the Disks of Active Galactic Nuclei*, 2021, ApJL, 923, L23

[12] **Wang Y.-H.**, Leigh N. W. C., Liu B., Perna R., *SpaceHub: A high-performance gravity integration toolkit for few-body problems in astrophysics*, 2021, MNRAS, 505, 1053

[11] **Wang Y.-H.**, Perna R., Armitage P. J., *Partial tidal disruption events by stellar mass black holes: Gravitational instability of stream and impact from remnant core*, 2021, MNRAS, 503, 6005

[10] **Wang Y.-H.**, Leigh N. W. C., Perna R., Shara M. M., *Hot Jupiter and ultra-cold Saturn formation in dense star clusters*, 2020, ApJ, 905, 136

[9] **Wang Y.-H.**, Perna R., Leigh N. W. C., *Planetary architectures in interacting stellar environments*, 2020, MNRAS, 496, 1453

[8] **Wang Y.-H.**, Perna R., Leigh N. W. C., *Giant Planet Swaps during Close Stellar Encounters*, 2020, ApJL, 891, L14

[7] **Wang Y.-H.**, Leigh N. W. C., Sesana A., Perna R., *The cosmological distribution of compact object mergers from dynamical interactions with SMBH binaries*, 2019, MNRAS, 490, 2627

[6] Liu B., Lai D., **Wang Y.-H.**, *Binary Mergers near a Supermassive Black Hole: Relativistic Effects in Triples*, 2019, ApJL, 883, L7

[5] Liu B., Lai D., **Wang Y.-H.**, *Black Hole and Neutron Star Binary Mergers in Triple Systems. II. Merger Eccentricity and Spin-Orbit Misalignment*, 2019, ApJ, 881, 41

[4] Perna R., **Wang Y.-H.**, Farr W. M., Leigh N., Cantiello M., *Constraining the Black Hole Initial Mass Function with LIGO/Virgo Observations*, 2019, ApJL, 878, L1

[3] **Wang Y.-H.**, Leigh N., Sesana A., Perna R., *Hypervelocity binaries from close encounters with a SMBH-IMBH binary: orbital properties and diagnostics*, 2019, MNRAS, 482, 3206

[2] **Wang Y.-H.**, Leigh N., Yuan Y.-F., Perna R., *The fate of close encounters between binary stars and binary supermassive black holes*, 2018, MNRAS, 475, 4595

[1] Liu B., **Wang Y.-H.**, Yuan Y.-F., *Modified evolution of stellar binaries from supermassive black hole binaries*, 2017, MNRAS, 466, 3376

SELECTED TALKS

Invited Talks & Seminars

- Transient Phenomena and Physical Processes Around Supermassive Black Holes — Oct. 2024
- Graduate Seminar, Nanjing University — Jun. 2023
- AGN Santa Fe Conference, Los Alamos National Lab — Mar. 2023
- Graduate Seminar, Stony Brook University — Feb. 2023
- Astro-coffee, IAS, Princeton University — Feb. 2023
- Bahcall Lunch Talk, IAS, Princeton University — Feb. 2023
- Graduate Seminar, Georgia Institute of Technology — Sep. 2022
- Astrophysics Seminar, University of the Balearic Islands — Oct. 2021
- Astronomy Seminar, Universidad de Concepción — Sep. 2021
- Astronomy Group Meeting, Cornell University — Jun. 2018

Contributed Talks

- 50 years of Binaries and Disks: Lubow@75 — Apr. 2024
- Anticipating the Rising Tide of Tidal Disruption Events, KITP — Apr. 2024
- 53rd Annual DDA Meeting, CCA, Flatiron Institute — Apr. 2022
- Planet Formation Group Meeting, CCA, Flatiron Institute — Oct. 2021
- Astronomy Seminar, Stony Brook University — Aug. 2021
- Compact Object Group Meeting, CCA, Flatiron Institute — Mar. 2021
- Astronomy Seminar, American Museum of Natural History — Aug. 2018

SOFTWARE & RESEARCH PRODUCTS

VegasAfterglow [GitHub]

Primary Developer | High-speed modeling engine for Gamma-Ray Burst (GRB) afterglows.

- Optimized C++ engine capable of generating light curves in milliseconds, enabling real-time data fitting and massive population studies.
- Supports diverse jet structures and emission mechanisms (Synchrotron & Inverse Compton) to model complex, realistic burst environments.
- Features a Python interface designed for seamless integration with standard fitting tools (like MCMC), allowing researchers to easily analyze observational data.
- *Publication:* Wang et al. (2025), JHEAP [Paper 30].

SpaceHub [GitHub]

Primary Developer | High-precision few-body gravity integration toolkit.

- Implemented Algorithmic Regularization (AR) and symplectic integrators to handle extreme mass ratios and eccentricities (e.g., SMBHs, chaotic star clusters).
- Designed an easy-to-use API allowing community adoption for diverse dynamical problems.
- *Publication:* Wang et al. (2021), MNRAS [Paper 12].

Secular [GitHub]

Primary Developer | Fast solver for hierarchical secular dynamics.

- Solves orbit-averaged equations of motion for hierarchical triples/quadruples.
- Highly optimized for studying long-term Lidov-Kozai oscillations and stability limits.

Code Extensions & Modifications

- **Phantom (SPH):** Implemented custom radiative Equation of State (EoS) and modified accretion capture logic to simulate Tidal Disruption Events (TDEs).
- **Athena++ (MHD):** Configured relativistic MHD setups for simulating short GRB jets and their interaction with the ambient medium.

TECHNICAL SKILLS

Languages: C/C++, Fortran, Python, LaTeX, HTML, Markdown
HPC: MPI, OpenMP, Slurm workload manager
Astrophysics Codes: Phantom, Athena++, Rebound, SpaceHub

MENTORING

Graduate Students

Connery Chen, University of Nevada, Las Vegas — 2023 - Present
Chaitanya Prasad, Stony Brook University — 2022 - 2023
Michael Ray, Stony Brook University — 2021 - 2022

Undergraduate Students

Antonio Frigo, Stony Brook University — 2021
Robert Serrano, Stony Brook University — 2019 - 2020

PROFESSIONAL SERVICE & OUTREACH

Collaboration Memberships

- Member, *Einstein Probe (EP)* Collaboration — 2024 - Present
- Member, *SVOM* (Space-based multi-band astronomical Variable Objects Monitor) — 2024 - Present

Referee Service: *ApJ*, *ApJL*, *MNRAS*, *A&A*, *JHEAp*.

Leadership:

- Coordinator, NCfA Multi-messenger Meeting 2022 - 2025
- Mentor, Stony Brook Graduate Mentoring Program 2017 - 2019

Outreach:

- UNLV Community Engagement Expo 2023
- Astronomy on Tap (Las Vegas): "100 Years of Variable Stars" 2023

TEACHING EXPERIENCE

Teaching Assistant, AST-203: *Astronomy* Spring 2019
Teaching Assistant, AST-248: *Search for Life in the Universe* Fall 2018
Teaching Assistant, PHY-134: *Classical Physics Lab II* Spring 2018
Teaching Assistant, AST-205: *Introduction to Planetary Sciences* Fall 2017
Teaching Assistant, *Electromagnetism* Fall 2014

SELECTED PRESS COVERAGE

Evidence of a Past Merger of the Galactic Center Black Hole

- *CBC:* "A cosmic collision 9 billion years ago..." Sept. 2024
- *Phys.org:* "Massive merger: Study reveals evidence..." Sept. 2024

Floating binary planets from ejections during close stellar encounters

- *Quanta Magazine:* "Rogue Worlds Throw Planetary Ideas Out of Orbit" Nov. 2023
- *Physics World:* "Pairs of rogue planets found wandering..." Oct. 2023

A planet could have been stolen from the solar system as it formed

- *New Scientist* Feb. 2020

REFERENCES

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